Program 3 : Code a Dockerized Python Flask or Nodejs Application

Project Structure

```
.
├── app.py
├── Dockerfile
├── __pycache__
├── requirements.txt
└── venv
```

STEP 1: Create a **Dockerfile** and add the following content.

```
flask1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab1$ cat Dockerfile
FROM python:3.9-slim

WORKDIR /app

LABEL maintainer="jayeshkumarr"

COPY requirements.txt .

RUN pip install -r requirements.txt

COPY . .

ENV FLASK_APP=app.py
ENV FLASK_RUN_HOST=0.0.0.0
ENV FLASK_RUN_PORT=5000

EXPOSE 5000

CMD ["flask", "run"]1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab1$
```

STEP 2: Create a Python file -**app.py**

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab1$ cat app.py
from flask import Flask

app = Flask(__name__)

@app.route('/')
def home():
    return "Hello Deepkia Ma'am"

if __name__ == '__main__':
    app.run(host="0.0.0.0.0", port=5000)
```

STEP 3: Create a simple text file – requirements.txt & add the following

Flask

STEP 4: Execute the Docker Build command on the Terminal to build the Docker image and then execute the Docker run command specifying the Port Number for the container.

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab1$ dockerscript
Enter the Name of the Image you want to create:
lab1
Image lab1 does not exist. Building it now...
[+] Building 13.7s (10/10) FINISHED                                                     docker:default
 => [internal] load build definition from Dockerfile                                              0.0s
 => => transferring dockerfile: 286B                                                              0.0s
 => [internal] load metadata for docker.io/library/python:3.9-slim                                4.0s
 => [internal] load .dockerignore                                                                 0.0s
 => => transferring context: 2B                                                                   0.0s
 => [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f14          3.1s
 => => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f14          0.0s
 => => sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3d 1.29MB / 1.29MB          0.9s
 => => sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710af 13.88MB / 13.88MB          1.9s
 => => sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c2c450078315beb5 251B / 251B          1.3s
 => => sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13c 10.36kB / 10.36kB          0.0s
 => => sha256:dad5b29e3506c35e0fd222736f4d4ef25d21b219acdd73f7bb41d59996 1.74kB / 1.74kB          0.0s
 => => sha256:085da638e1b8a449514c3fda83ff50a3bffae4418b050cfacd87e57220 5.40kB / 5.40kB          0.0s
 => => extracting sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881          0.1s
 => => extracting sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c0          0.9s
 => => extracting sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c2c450078315beb50          0.0s
 => [internal] load build context                                                                 0.0s
 => => transferring context: 93B                                                                  0.0s
```

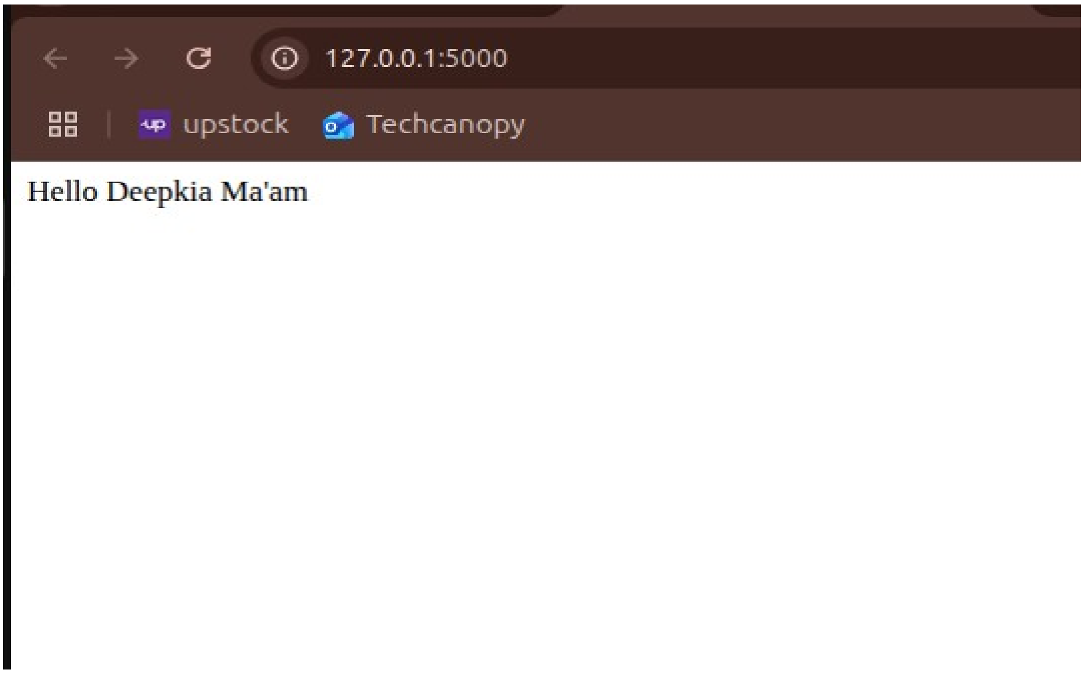
```
Terminal
 => => writing image sha256:65946050b098ead64dc0884c50b7562d73f678825b6d16cc2a37356f334a          0.0s
 => => naming to docker.io/library/lab1                                                           0.0s

Do you want to run the Docker container now? (y/n):
y
Enter the number of flags you want to add to 'docker run':
3
Enter flag #1 (e.g., -p 8080:80 or --name mycontainer):
-p 5000:5000
Enter flag #2 (e.g., -p 8080:80 or --name mycontainer):
--name lab1_cont
Enter flag #3 (e.g., -p 8080:80 or --name mycontainer):
-it

Enter the container name (optional, press Enter to skip):


Running command: sudo docker run  -p 5000:5000 --name lab1_cont -it lab1
 * Serving Flask app 'app.py'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a producti
on WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
```

STEP 5: Test the Container by verifying the localhost details on web browser, the text



STEP 6: Stop the container with the container number, remove the docker container and delete the Docker Image with the name program-3

sudo docker container stop lab3_cont
sudo docker container rm lab3_cont
sudo docker image rm lab3